LINUX COMMANDS

SYSTEM COMMANDS

`uname`

- This command displays basic information about the operating system.
  It helps you identify which OS is running on your machine.
  It is commonly used for troubleshooting and system verification.

`uname -r`

- This shows the Linux kernel version currently running.
  Kernel is the core part of the OS managing hardware and processes.
  Useful for compatibility checks and updates.

`uname -a`

- This displays all system information including kernel, hostname, and architecture.
  It gives a complete overview of the system in one command.
  Mostly used for debugging and system auditing.

`uptime`

- This shows how long the system has been running.
  It also displays load average and active users.
  Useful for monitoring system stability.

`hostname`

- This prints the system’s hostname.
  Hostname is the name assigned to a machine in a network.
  It helps identify systems in servers or cloud environments.

HARDWARE COMMANDS

`lscpu`

- This command displays CPU architecture details.
  It shows cores, threads, and processor type.
  Useful for performance analysis and system tuning.

`free -m`

- This shows memory usage in MB.
  It displays total, used, and free RAM.
  Helpful for monitoring system memory.

`df -h`

- This shows disk space usage in human-readable format.
  It displays available and used storage for file systems.
  Useful for checking disk capacity.

`lsblk -a`

- This lists all block devices like disks and partitions.
  It shows mount points and device structure.
  Helpful for storage management.

FILE COMMANDS

`touch file`

- This creates an empty file.
  If the file exists, it updates the timestamp.
  Used for quick file creation.

`rm file`

- This deletes a file.
  It permanently removes the file from the system.
  Use carefully as it cannot be undone.

`mkdir folder`

- This creates a new directory.
  It helps organize files into folders.
  Commonly used in project setup.

`cd folder`

- This changes the current directory.
  It allows navigation between folders.
  Essential for working in Linux filesystem.

`pwd`

- This shows the present working directory.
  It tells your current location in the system.
  Useful for navigation clarity.

USER COMMANDS

`useradd username`

- This creates a new user in the system.
  It assigns default settings like home directory.
  Used in system administration.

`userdel username`

- This deletes a user account.
  It removes user access from the system.
  Can be forced using options.

`passwd username`

- This sets or changes a user password.
  It ensures secure authentication.
  Essential for user management.

PERMISSION COMMANDS

`chmod 777 file`

- This gives full permissions to everyone.
  It allows read, write, and execute access.
  Not recommended for secure environments.

`chown user file`

- This changes the file owner.
  It assigns ownership to a specific user.
  Used in permission management.

`chgrp group file`

- This changes the group of a file.
  It helps manage access for multiple users.
  Used in team-based environments.

File Discovery Commands

`find . -name file`

- This searches for a file in the current directory.
  It can locate files based on name or conditions.
  Very powerful for deep searches.

`locate filename`

- This quickly finds files using a database.
  It is faster than find but may need database updates.
  Useful for quick file lookup.

`grep "word" file`

- This searches for a word inside a file.
  It prints lines containing the match.
  Widely used for log analysis.

File & Directory

`cp`

- This command is used to copy files or directories.
  It helps duplicate data from one location to another.
  It supports recursive copying using `-r`.

`mv`

- This command is used to move or rename files.
  It can change file location or rename in one step.
  Very commonly used in file management.

`stat`

- This shows detailed information about a file.
  It includes size, permissions, and timestamps.
  Useful for debugging file issues.

Text Processing

`awk`

- This is used for pattern scanning and processing text.
  It can extract columns and perform calculations.
  Widely used in logs and data parsing.

`sed`

- This is a stream editor used to modify text.
  It can replace, delete, or update content in files.
  Useful for automation scripts.

`sort`

- This sorts lines in a file.
  It can sort alphabetically or numerically.
  Helpful for organizing data.

`uniq`

- This removes duplicate lines from input.
  It works with sorted data.
  Useful for data cleanup.

Process Management

`kill`

- This command is used to terminate processes.
  It sends signals to stop running programs.
  Useful when a process is stuck.

`kill -9`

- This forcefully stops a process.
  It cannot be ignored by the process.
  Use carefully as it may cause data loss.

`ps -ef`

- This shows all running processes.
  It provides detailed process information.
  Useful for monitoring.

Networking

`ping`

- This checks network connectivity.
  It sends packets to a remote host.
  Useful for troubleshooting network issues.

`netstat`

- This shows network connections and ports.
  It helps identify open ports and services.
  Useful for debugging servers.

`curl`

- This is used to transfer data from URLs.
  It helps test APIs and endpoints.
  Very important in DevOps.

`wget`

- This downloads files from the internet.
  It supports background downloads.
  Useful for automation scripts.

Permissions & Ownership

`umask`

- This defines default file permissions.
  It controls permission creation rules.
  Important for security.